

Windows Server 2008

From MS Official Course Content

Module 1: Installing and Configuring Servers

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Module Overview

- Installing Windows Server 2008
- Managing Server Roles and Features
- Overview of the Server Core Installation Option

Lesson 1: Installing Windows Server 2008

- Windows Server 2008 Editions
- Windows Server 2008 Installation Requirements
- x64 Installation Considerations
- Common Installation Scenarios
- Preparing for the Installation of Windows Server 2008
- Process for Installing Windows Server 2008

Windows Server 2008 Editions

Edition	Server Core
Windows Server 2008 Standard	Yes
Windows Server 2008 Enterprise	Yes
Windows Server 2008 DataCenter	Yes
Windows Web Server 2008	No
Windows Server 2008 for Itanium-based Systems	No

Windows Server 2008 Installation Requirements

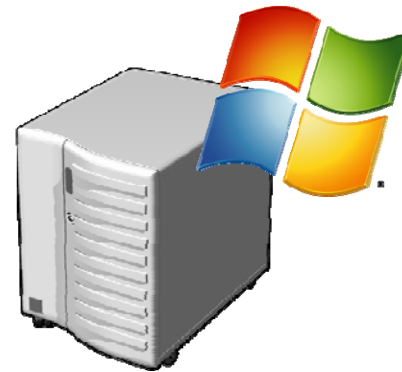
Component	Requirement
Processor	• Minimum: 1 Ghz • Recommended: 2 Ghz • Optimal: 3 Ghz or faster
Memory	• Minimum: 512 MB RAM • Recommended: 1 GB RAM • Optimal: 2 GB RAM (Full) or 1 GB RAM (Server Core) or more • Maximum (32-bit): 4 GB (Standard) or 64 GB (Enterprise and Datacenter) • Maximum (64-bit): 32 GB (Standard) or 2 TB (Enterprise, Datacenter, and Itanium-based systems)
Available Disk Space	• Minimum: 8 GB • Recommended: 40 GB (Full); 10 GB (Core) • Optimal: 80 GB (Full); 40 GB (Core)
Optical Drive	• DVD-ROM
Display and Peripherals	• Super VGA (800 x 600) or higher-resolution monitor • Keyboard • Microsoft mouse or compatible pointing device

x64 Installation Considerations

- **Requires that all kernel-mode device drivers are digitally signed**
- **Device drivers are necessarily trusted because they have direct access to the system's hardware**
- **May be able to mitigate some “rootkit” software**

Common Installation Scenarios

- **Clean installations**
- **Upgrades**
- **Unattended installations**



Preparing for the Installation of Windows Server 2008

- ✓ Check for application compatibility
- ✓ Disconnect UPS Devices
- ✓ Back up existing servers to be upgraded
- ✓ Disable virus protection
- ✓ Run the Windows Memory Diagnostic Tool
- ✓ Obtain mass storage drivers
- ✓ Be aware of Windows Firewall installation defaults
- ✓ Prepare Active Directory for the Active Directory Domain Services role

Process for Installing Windows Server 2008

- 1** Provide language and preferences
- 2** Setup copies and installs Windows
- 3** Administrator configures the server using the Initial Configuration Tasks page at first logon

Lesson 2: Managing Server Roles and Features

- Tools Used for Administrative Tasks
- What Are Server Roles?
- What Are Server Features?
- Demonstration: Installing Server Roles and Features Using Server Manager

Tools Used for Administrative Tasks

The screenshot shows the 'Initial Configuration Tasks' window in Windows Server. The window has a title bar with the text 'Initial Configuration Tasks' and standard window controls. Below the title bar is a header area with a wrench icon, the text 'Perform the following tasks to initially configure this server', and the Windows Server logo with 'Code Name "Longhorn"'. The main content area is divided into three sections, each with a numbered icon and a title. Section 1, 'Provide Computer Information', includes tasks for setting the time zone, configuring networking, and providing computer name and domain. Section 2, 'Update This Server', includes tasks for enabling automatic updates and downloading/installing updates. Section 3, 'Customize This Server', includes tasks for adding roles, adding features, enabling Remote Desktop, and configuring Windows Firewall. Each task is accompanied by an icon and a description of the current configuration. At the bottom of the window, there is a checkbox for 'Do not show this window at logon' and a 'Close' button.

Initial Configuration Tasks

Perform the following tasks to initially configure this server

1 Provide Computer Information Specifying computer information

- Set time zone** **Time Zone:** (GMT-08:00) Pacific Time (US Canada)
- Configure networking** **Local Area Connection:** IPv4 address assigned by DHCP, IPv6 enabled
- Provide computer name and domain** **Full Computer Name:** WIN-28QJ9SC8T38
Workgroup: WORKGROUP

2 Update This Server Updating your Windows server

- Enable automatic updating and feedback** **Updates:** Not configured
Feedback: Windows Error Reporting on
Participating in Customer Experience Improvement Program
- Download and install updates** **Checked for Updates:** Never
Installed Updates: Never

3 Customize This Server Customizing your server

- Add roles** **Roles:** None
- Add features** **Features:** Remote Differential Compression
- Enable Remote Desktop** **Remote Desktop:** Disabled
- Configure Windows Firewall** **Firewall:** On

[Print, e-mail, or save this information](#)

☐ Do not show this window at logon Close

What Are Server Roles?

A server role describes the primary function of a server

- **Administrators can dedicate an entire computer to one role or install multiple server roles on a single computer**
- **Each role can include one or more role services or subelement of a role**
- **Server Manager is the tool that is used to install, configure, and remove Server Roles**



What Are Server Features?

Server features provide auxiliary or supporting functions to servers

- **Typically, administrators add features, not as the primary function of a server, but to augment the functionality of installed roles**
- **Server Manager is the tool that you can use to install, configure, and manage features in Windows Server 2008**



Demonstration: Installing Server Roles and Features Using Server Manager

In this demonstration, you will see how to use Server Manager to install roles and features

Lesson 3: Overview of the Server Core Installation Option

- Benefits of a Server Core Installation
- Server Roles That a Server Core Installation Supports
- Features That a Server Core Installation Supports
- Managing a Server Core Installation

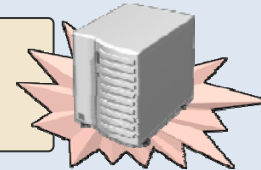
Benefits of a Server Core Installation

Benefits of a Server Core installation include:



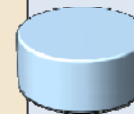
- **Reduced maintenance**

- **Reduced attack surface**



- **Reduced management**

- **Less disk space required**



Server Roles That a Server Core Installation Supports

A Server Core installation supports the following server roles:

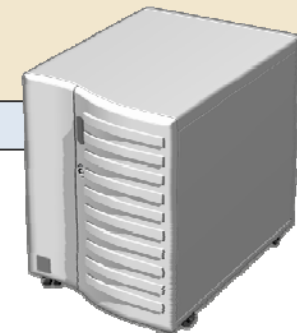
- Active Directory Domain Services
- Active Directory Lightweight Directory Services
- DHCP Server
- DNS Server
- Web Server (IIS)
- File Services
- Print Server
- Streaming Media Services
- Windows Server Virtualization (Hyper-V)



Features That a Server Core Installation Supports

A Server Core installation supports the following features:

- Backup
- Bitlocker Drive Encryption
- Failover Clustering
- Multitpath I/O
- Network Load Balancing
- Removable Storage
- Simple Network Management Protocol
- Subsystem for UNIX-based applications
- Telnet Client
- WINS



Managing a Server Core Installation

You can manager a Server Core installation:

- **Locally and remotely using a command prompt**
- **Remotely using Terminal Server**
- **Remotely using Windows Remote Shell**
- **Remotely using an MMC snap-in**

Lab Review

- What must you consider when choosing between a Standard and a Core installation of Windows Server 2008?
- Can you install Windows Server 2008 without specifying a product key during installation? If so, what are some of the disadvantages?
- How would you configure a firewall to allow Terminal Services connections on a Server Core installation?

Module Review and Takeaways

- Review Questions
- Considerations for Installing Windows Server 2008
- Best Practices
- Tools